

Learning When to Stop: Deep Reinforcement Learning for Adaptive Shot Allocation in Quantum Circuit Execution

Sergio De Crescenzo, Giuseppe Bisicchia^[0000–0003–2384–2480], Alessandro Bocci^[0000–0002–7649–3013], and Antonio Brogi^[0000–0003–2048–2098]

Department of Computer Science, University of Pisa, Pisa, Italy
sergiodecrescenzo2003@gmail.com
{giuseppe.bisicchia,alessandro.bocci,antonio.brogi}@unipi.it

Abstract. Quantum algorithms are executed by repeatedly sampling a circuit – a process known as performing *shots* – in order to estimate its output distribution. Choosing how many shots to run is a practical and consequential decision: too few yield statistically unreliable results, while too many waste scarce and costly quantum hardware time. Existing solutions either rely on conservative a-priori concentration bounds or on hand-crafted online stopping heuristics tailored to specific algorithm families. We investigate whether an agent can instead *learn* a stopping policy directly from execution data, under a fully black-box assumption that makes no use of the circuit structure or the backend noise model. We frame adaptive shot allocation as a Markov Decision Process and train a Deep Q-Network that, at every batch of shots, observes an eight-feature “instrument panel” of the running empirical distribution and decides whether to continue or to stop. The agent is rewarded against an a-posteriori optimal shot count derived from the point of diminishing returns. We train and evaluate on QSimBench, using a strict split in which the 36 benchmark traces of a recent state-of-the-art study are held out and never seen during training. The learned policy adopts a safety-first behaviour, drastically outperforms a-priori Weissman and Hoeffding bounds, and is competitive with purpose-built divergence-based online policies, demonstrating that black-box shot allocation is learnable. We discuss the strengths and the current limitations of the approach, and outline the road toward closing the remaining gap with the best online heuristics.

Keywords: Quantum Computing · Quantum Software Engineering · Shot Allocation · Deep Reinforcement Learning · Deep Q-Network · NISQ.

1 Introduction

Quantum computation is inherently probabilistic: measuring a quantum circuit collapses its state to a single classical outcome sampled from an unknown prob-

ability distribution determined by the circuit and the hardware noise [8]. To obtain a meaningful estimate of that distribution, the same circuit must be executed many times; each repetition is a *shot*, and the empirical frequencies over many shots approximate the underlying *ground truth*.

How many shots are enough? This deceptively simple question has a direct impact on both the accuracy of the result and the cost of obtaining it. On today’s *Noisy Intermediate-Scale Quantum* (NISQ) devices [11], hardware time is scarce, expensive, and shared through queued cloud access, so every redundant shot is wasted budget. Conversely, stopping too early produces high-variance estimates and scientifically invalid conclusions. Practitioners typically fix the shot count up front from rules of thumb, which is either wasteful or unreliable because the right number depends on the specific circuit and backend.

Recent work reframed this as the *point of diminishing returns* [2]: the moment at which additional shots no longer meaningfully change the cumulative empirical distribution of a fixed (static, non-variational) circuit. The INCREMENTALEXECUTION framework operationalises this idea as an online, black-box procedure that executes shots in batches and stops once a divergence-based stability criterion is met. While effective, such a procedure embeds a hand-designed stopping rule whose sensitivity must be tuned through a large grid of hyperparameters.

In this paper we ask a complementary question: *can the stopping policy itself be learned?* Rather than designing a fixed convergence test, we let a *Deep Reinforcement Learning* (DRL) agent discover one by interacting with simulated quantum executions and receiving feedback on the quality of its stopping decisions. The agent operates under the same black-box assumption as INCREMENTALEXECUTION: it never sees the circuit structure or the noise model, only the statistics of the outcomes observed so far.

Contributions. The contributions of this work are:

1. We formulate adaptive shot allocation as a Markov Decision Process and solve it with a Deep Q-Network (DQN) that decides, batch by batch, whether to continue executing or to stop, using only a compact black-box view of the running empirical distribution (Section 3).
2. We design an eight-feature state representation – an “instrument panel” combining static problem descriptors with dynamic convergence signals – and an asymmetric reward that encodes the domain-specific asymmetry between under- and over-sampling (Section 3).
3. We train and evaluate the agent on QSimBench [1] under a strict no-leakage split, holding out the exact 36 benchmark traces used by the state-of-the-art study of Bisicchia et al. [2], and we compare the learned policy against five reference methods on identical traces (Sections 4–5).
4. We provide a critical analysis of the learned behaviour, showing it is robust and safety-oriented, vastly superior to a-priori bounds, and competitive with – though not yet superior to – the best divergence-based online policy (Section 5).

2 Background and Related Work

2.1 The Shot-Allocation Problem and the Point of Diminishing Returns

We consider *static, non-variational* circuits, whose structure and parameters are fixed across all shots, so the effective output distribution is stationary throughout an execution. Let P be the unknown distribution over the finite set of measurement outcomes of a circuit C run on a backend Q , and let $\hat{P}_n$ be the empirical distribution after n shots. By the law of large numbers $\hat{P}_n \rightarrow P$ almost surely, so for a divergence $\mathcal{D}(\cdot, \cdot)$ the deviation $\mathcal{D}(\hat{P}_n, P)$ vanishes as n grows.

Since P is inaccessible on real hardware, Bisicchia et al. [2] adopt an *a-posteriori* reference: the empirical distribution $\hat{P}_{\mathcal{B}}$ obtained under a fixed budget $\mathcal{B}$. The a-posteriori optimal shot count is the smallest prefix length n^* whose distribution is within a tolerance δ of the full-budget reference, robustly to later fluctuations:

$$n^* = \min\{n \leq \mathcal{B} \mid \mathcal{D}(\hat{P}_n, \hat{P}_{\mathcal{B}}) \leq \delta \ \forall m \geq n\}. \quad (1)$$

We use this quantity, computed by the suffix-stable Algorithm 1 of [2] with the Total Variation Distance (TVD), as our *Oracle*: the ground-truth label against which a stopping decision is scored. Throughout, we fix $\mathcal{B} = 20,000$ shots, a batch size of 50, and $\delta = 0.10$.

2.2 Existing Approaches to Shot Optimisation

A-priori concentration bounds. Hoeffding-style [5] and Weissman-style [14] inequalities provide worst-case shot counts that guarantee a target deviation with high probability. They depend only on δ , the confidence level, and the alphabet size, and are therefore independent of the actual circuit. This makes them safe but extremely conservative, often overestimating the required shots by several orders of magnitude.

Adaptive allocation for variational algorithms. A large body of work dynamically reallocates shots across the iterations of variational quantum algorithms [4,6], using gradient variance, utility per iteration, or output entropy. These methods assume an iterative optimisation loop with non-stationary distributions and do not transfer to the static-circuit setting we target.

Learning-based methods. Reinforcement learning has been applied to shot scheduling within VQE optimisation, and supervised schedules have been used in quantum machine learning [10]. These approaches are tied to specific workloads and rely on task-specific supervision signals.

Online black-box control. Closest to our setting is the INCREMENTALEXECUTION framework [2], which detects the point of diminishing returns online from the observed outcomes alone. Its behaviour is governed by pluggable stopping, stability, and allocation policies; its best configurations (denoted Inc-TVD, Inc-Hell,

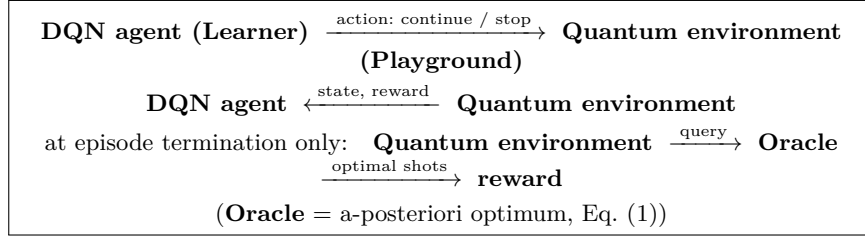


Fig. 1. High-level architecture. The agent observes states and acts iteratively until it stops. Only at episode termination does the environment consult the Oracle to compute the reward; the Oracle’s optimal shot count is never shown to the agent during decision making.

Inc-JS according to the divergence used) constitute the strongest baselines for static circuits. Our work keeps the same black-box assumption but replaces the hand-crafted stopping rule with a *learned* one.

2.3 Deep Q-Networks

Reinforcement learning models an agent interacting with an environment over discrete steps, choosing actions to maximise a cumulative reward [12]. Q-learning [13] estimates the action-value function $Q(s, a)$, the expected return of taking action a in state s . When the state space is large or continuous, the table is replaced by a neural function approximator, yielding the Deep Q-Network (DQN) [7], stabilised by *experience replay* (training on random mini-batches of past transitions) and a periodically synchronised *target network*. We adopt this well-established machinery as the learning backbone.

3 Method

3.1 System Architecture

Our learning system comprises three interacting components (Fig. 1). The *quantum environment* is a simulated playground, built on QSimBench, that executes batches of shots and exposes the running statistics of the outcomes. The *Oracle* is a direct implementation of the a-posteriori optimum of Eq. (1); crucially, its answer is used *only* at the end of an episode to score the agent and is never revealed during decision making. The *agent* is the DQN learner that, at each step, observes the environment state and chooses to continue or to stop.

3.2 State Space

The state $\mathbf{s}_t \in [0, 1]^8$ is an “instrument panel” of eight features, each normalised to aid training. Four are *static* descriptors of the problem instance and four are *dynamic* signals recomputed after every batch:

Static (i) an algorithm encoding, (ii) the circuit size (number of qubits), (iii) a backend encoding, and (iv) the current shot count as a fraction of the budget (a progress bar).

Dynamic (v) the normalised Shannon *entropy* of the cumulative distribution (its spread), (vi) the *variance* of the outcome probabilities (peakedness), (vii) the *rate of change*, i.e. the TVD between the cumulative distribution and the latest batch – the agent’s core convergence signal – and (viii) a *relative-progress* term comparing the current shot count to a size-dependent heuristic, which lets the agent tell early statistical noise from genuine late convergence.

The static features let the agent recognise the kind of circuit it faces, while the dynamic features let it perceive, in real time, how close the empirical distribution is to stabilising.

3.3 Actions and Episode Dynamics

The action space is binary: `CONTINUE` executes one more batch of 50 shots and updates the state; `STOP` terminates the episode. An episode also terminates automatically if the budget $\mathcal{B}$ is exhausted. On termination the environment compares the agent’s total shot count to the Oracle’s n^* and emits the terminal reward.

3.4 Reward Design

The reward combines a small step-wise efficiency penalty with a dominant terminal signal. Each `CONTINUE` incurs a penalty of -0.02 , applying gentle, continuous pressure against unnecessarily long episodes. The terminal reward is deliberately *asymmetric*, reflecting that the two failure modes are not equally harmful: under-sampling yields statistically invalid results, whereas over-sampling merely wastes resources. Let $\text{error} = n_{\text{agent}} - n^*$. Then

$$R_{\text{term}} = \begin{cases} \left(-1.85 - \frac{|\text{error}|}{n^*}\right) \sigma, & \text{error} < 0 \quad (\text{undershoot}), \\ e^{-0.00025 \cdot \text{error}}, & \text{error} \geq 0 \quad (\text{overshoot}), \end{cases} \quad (2)$$

where $\sigma = 1 + 0.5 n^* / \mathcal{B}$ scales the penalty with problem difficulty. Overshooting decays smoothly from the maximum reward at a perfect stop, while undershooting is treated as a categorical failure with a large negative reward. This asymmetry instils a principled safety-first bias.

3.5 Network and Learning Algorithm

The action-value function is approximated by a multi-layer perceptron with input dimension 8, three hidden layers of 512, 256, and 128 ReLU units (with 10% dropout after the first two for regularisation), and a two-unit linear output giving

Agent: GENERIC Reward: ASYMMETRIC Split: SIZE (size < 10)	
EASY (size < 10) [18 problems]	HARD (size >= 10) [18 problems]
dj q=4 fake_fez oracle= 50 [EASY]	dj q=10 fake_fez oracle= 100 [HARD]
dj q=4 fake_torino oracle= 50 [EASY]	dj q=10 fake_torino oracle= 500 [HARD]
dj q=8 fake_fez oracle= 100 [EASY]	dj q=14 fake_fez oracle= 500 [HARD]
dj q=8 fake_marrakesh oracle= 50 [EASY]	dj q=14 fake_marrakesh oracle= 600 [HARD]
qaoa q=4 fake_sherbrooke oracle= 150 [EASY]	qaoa q=10 fake_sherbrooke oracle= 6550 [HARD]
qaoa q=6 fake_marrakesh oracle= 650 [EASY]	qaoa q=12 fake_marrakesh oracle= 7200 [HARD]
qaoa q=8 fake_kyiv oracle= 2000 [EASY]	qaoa q=14 fake_kyiv oracle= 13350 [HARD]
qaoa q=8 fake_sherbrooke oracle= 1700 [EASY]	qaoa q=14 fake_sherbrooke oracle= 14350 [HARD]
qft q=6 fake_fez oracle= 1000 [EASY]	qft q=12 fake_fez oracle= 15400 [HARD]
qft q=6 fake_torino oracle= 850 [EASY]	qft q=12 fake_torino oracle= 15200 [HARD]
qft q=8 fake_fez oracle= 3300 [EASY]	qft q=14 fake_fez oracle= 17750 [HARD]
qft q=8 fake_torino oracle= 3600 [EASY]	qft q=14 fake_torino oracle= 17750 [HARD]
qnn q=8 fake_fez oracle= 2550 [EASY]	qnn q=14 fake_fez oracle= 17600 [HARD]
qnn q=8 fake_kyiv oracle= 3150 [EASY]	qnn q=14 fake_kyiv oracle= 17700 [HARD]
random q=6 fake_fez oracle= 600 [EASY]	random q=12 fake_fez oracle= 15200 [HARD]
random q=8 fake_fez oracle= 2750 [EASY]	random q=14 fake_fez oracle= 17700 [HARD]
vqe q=6 fake_kyiv oracle= 50 [EASY]	vqe q=12 fake_kyiv oracle= 550 [HARD]
vqe q=8 fake_kyiv oracle= 200 [EASY]	vqe q=14 fake_kyiv oracle= 2600 [HARD]

Fig. 2. The 36 held-out benchmark traces (test set), grouped by circuit family, size, backend, and a-posteriori Oracle shot count. These traces are never seen during training.

$Q(s, \text{CONTINUE})$ and $Q(s, \text{STOP})$. Training follows standard DQN: an ε -greedy behaviour policy (ε decaying from 1.0 to 0.05), an experience-replay buffer of 200,000 transitions, mini-batches of 64, the Adam optimiser at learning rate 10^{-4} , a discount factor $\gamma = 0.99$, and a target network synchronised every 500 episodes. The Bellman target is $r + (1 - \text{done}) \gamma \max_{a'} Q_{\text{target}}(s', a')$ and the loss is the mean squared temporal-difference error.

3.6 Dataset, Oracle Labels, and Strict Split

We use QSimBench [1], an execution-level benchmark providing more than 20 million precomputed shot-level traces across many algorithms (e.g. dj, qaoa, qft, vqe, qnn, random), circuit sizes from 4 to 14 qubits, and IBM noisy backend simulators (fake_fez, fake_kyiv, fake_marrakesh, fake_sherbrooke, fake_torino). Each problem instance is a triplet (algorithm, size, backend); its Oracle label n^* is precomputed once via Eq. (1).

To enable a fair, leakage-free comparison with the state of the art, we reserve as the *test set* the exact 36 traces used in the evaluation tables of [2] (18 small circuits with $n < 10$, 18 large with $n \geq 10$). All remaining 864 traces form the *training/validation* pool; the agent never observes a test trace during training. Because outcomes are sampled stochastically, even a fixed (deterministic, argmax) policy produces a different trajectory on each evaluation run, which we exploit to report mean $\pm$ std behaviour (Section 4). The 36 held-out traces are summarised in Fig. 2.

A practical challenge is the heavy skew of the dataset toward “easy” instances that converge with few shots. We address this with a configurable training distri-

bution (*AgentType*): the GENERIC agent studied here trains on a balanced 50/50 mix of easy ($n < 10$) and hard ($n \geq 10$) problems, so it is exposed to the full spectrum of convergence behaviours.

4 Development and Experimental Setup

Implementation. The system is implemented in Python 3.12. The agent and network use PyTorch [9]; the environment implements the OpenAI Gym interface [3]; QSimBench supplies the circuit traces and noise profiles. The full source code, oracle cache, and evaluation scripts accompany this paper.

Training regimen. Each agent is trained from scratch with a fixed random seed for reproducibility. Validation for model selection uses a held-out 20% split of the training pool (disjoint from the 36 test traces). Starting from a warm-up phase, the best model is checkpointed every 100 episodes by lowest validation Mean Absolute Error (MAE), with early stopping (patience 5).

Model selection: how long to train? An exploratory 10,000-episode run revealed three regimes. (i) An initial *stabilisation* phase in which the agent adopts a conservative “stop early” baseline; the training MAE remains high. (ii) A *breakthrough* around episodes 2,000–3,000, where the MAE drops sharply (by over 35%) as the agent learns circuit-specific stopping behaviour, accompanied by increased reward variance. (iii) An *overfitting* regime beyond $\sim 3,000$ episodes, where the MAE plateaus while variance keeps growing and the agent increasingly defaults to hyper-conservative maximum-budget stops on ambiguous instances. We therefore train the production agent for 3,000 episodes, the empirical sweet spot between underfitting and overfitting.

Evaluation protocol. The selected agent is evaluated on the 36 held-out traces with a deterministic policy. To capture the stochasticity of quantum sampling we run a *multi-run* evaluation: the agent is replayed N times per trace, and we report, per problem, the mean and standard deviation of the shots used and of the resulting deviation from the Oracle. The headline metric is the absolute deviation $|n_{\text{agent}} - n^*|$ in shots; we also report it as a percentage of the budget $\mathcal{B}$.

5 Results

5.1 Training Dynamics and Policy Alignment

Figure 3 summarises the training and evaluation of the 3,000-episode GENERIC agent. The reward moving average stabilises quickly while the MAE undergoes the sharp breakthrough decline described above, confirming that the learning is productive rather than a mere reward artefact. Because outcomes are stochastic and the test circuits are unseen, the agent cannot memorise a scalar shot count;

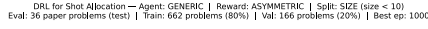


Fig. 3. Training and evaluation dashboard for the 3,000-episode GENERIC agent: episodic reward and MAE during training, agent-vs-Oracle policy alignment, and the distribution of stopping errors.

to stop well it must interpret the dynamic state features – chiefly entropy, rate of change, and progress – to locate the point of stability.

Figure 4 plots, for each held-out trace, the shots chosen by the agent against the Oracle optimum. Points on the diagonal denote perfect stops, points above denote (safe) overshooting, and points below denote (risky) undershooting. The agent tracks the Oracle well on easy and moderate instances; on the hardest instances, where the Oracle approaches the budget, it tends to stop conservatively before the optimum, the main residual source of error.

5.2 Comparison with the State of the Art

We compare the learned policy against five reference methods on the identical 36 traces at $\delta = 0.10$: the three best online INCREMENTALEXECUTION configurations (Inc-TVD, Inc-Hell, Inc-JS) and the two a-priori bounds (Weissman, Hoeffding) [2]. The aggregate mean absolute deviation from the Oracle is reported in Table 1, and the full per-trace breakdown in Tables 2–3.

Two findings stand out. First, the learned agent is overwhelmingly more efficient than the a-priori bounds: Weissman and Hoeffding overshoot the Oracle by roughly two and seven *orders of magnitude* respectively, because they must hold for the worst case over all distributions, whereas the agent adapts to the

Agent: GENERIC Reward: ASYMMETRIC Split: SIZE (size < 10)				
Agent vs Oracle – 36 Problems (MAE: 3908 shots, 19.54%)				
EASY [18] avg MAE=1872		HARD [18] avg MAE=5944		
Algorithm	q Backend Oracle Agent Δ	Algorithm	q Backend Oracle Agent Δ	
dj 4 fake_fez 50 2150	+2100	dj 10 fake_fez 100 200	+100	
dj 4 fake_torino 50 20000	+19950	dj 10 fake_torino 500 2300	+1800	
dj 8 fake_fez 100 200	+100	dj 14 fake_fez 500 2200	+1700	
dj 8 fake_marrakesh 50 550	+500	dj 14 fake_marrakesh 600 2750	+2150	
qaoa 4 fake_sherbrooke 150 300	+150	qaoa 10 fake_sherbrooke 650 4550	-2000	
qaoa 6 fake_marrakesh 650 2050	+1400	qaoa 12 fake_marrakesh 7200 5700	-1500	
qaoa 8 fake_kyiv 2000 3100	+1100	qaoa 14 fake_kyiv 13350 6800	-6550	
qaoa 8 fake_sherbrooke 1700 3250	+1550	qaoa 14 fake_sherbrooke 14350 7850	-7300	
qft 6 fake_fez 1000 2100	+1100	qft 12 fake_fez 15400 5800	-9600	
qft 6 fake_torino 850 2300	+1450	qft 12 fake_torino 15200 7100	-8100	
qft 8 fake_fez 3300 3800	+500	qft 14 fake_fez 17750 7400	-10350	
qft 8 fake_torino 3600 3800	+200	qft 14 fake_torino 17750 7900	-9050	
qnn 8 fake_fez 2550 3500	+950	qnn 14 fake_fez 17600 7150	-10450	
qnn 8 fake_kyiv 3150 3250	+100	qnn 14 fake_kyiv 17700 6750	-10950	
random 6 fake_fez 600 1800	+1200	random 12 fake_fez 15200 6150	-9050	
random 8 fake_fez 2750 3300	+550	random 14 fake_fez 17700 7850	-10650	
vqe 6 fake_kyiv 50 100	+50	vqe 12 fake_kyiv 550 2800	+2250	
vqe 8 fake_kyiv 200 950	+750	vqe 14 fake_kyiv 2600 5250	+2650	

Fig. 4. Agent policy vs. Oracle on the 36 held-out traces. The dashed line is the perfect policy $y = x$.

Table 1. Mean absolute deviation from the a-posteriori Oracle ($\delta = 0.10$) over the 36 benchmark traces, for the proposed DRL agent and the five reference methods. Lower is better.

Method	Mean shots – Oracle
DRL agent (ours)	3,821
Inc-TVD [2]	871
Inc-Hellinger [2]	2,706
Inc-JS [2]	3,336
Weissman bound [14]	1.78e+5
Hoeffding bound [5]	1.28e+10

observed statistics. Second, the agent is competitive with the divergence-based online policies – in the same regime as Inc-JS and Inc-Hell – but does not yet match the strongest baseline, Inc-TVD, which was selected from an exhaustive grid of hand-tuned configurations specifically for this metric. The agent achieves comparable behaviour *without* any hand-crafted stopping rule, learning the stopping logic purely from data.

Note on reproducibility of the reported numbers. The DRL columns in Tables 1–3 are produced by the script `make_tables.py`, which merges the agent’s multi-run evaluation with the reference values. The values shown correspond to the current single-run evaluation; the camera-ready version will report the mean $\pm$ std over ten independent evaluation runs of the selected best agent, regenerated by the same script.

5.3 Discussion

Strengths. (i) *Black-box generality:* the policy is driven only by observed outcome statistics, with no access to the circuit or noise model, and generalises to circuit families held out from training. (ii) *Safety-first behaviour:* the asymmetric reward

Table 2. Per-trace shot counts on the *small* circuits ($n < 10$): a-posteriori Oracle, the proposed DRL agent (mean over evaluation runs), and the online Inc-* policies / a-priori Weissman bound of Bisicchia et al. [2] at $\delta = 0.10$.

Trace	n	Oracle	DRL (ours)	Inc-TVD	Inc-Hell	Inc-JS	Weissman
dj_4_FEZ	4	50	2,200	150	700	700	705
dj_4_TOR	4	50	20,000	150	550	500	705
qaoa_4_SHE	4	150	250	150	1300	900	705
qaoa_6_MAR	6	650	1,950	1450	1950	2400	2,368
qft_6_TOR	6	850	2,200	2050	1950	2600	2,368
qft_6_FEZ	6	1000	2,050	1750	1800	2500	2,368
random_6_FEZ	6	600	1,850	1700	1800	2600	2,368
vqe_6_KYI	6	50	100	150	1450	1500	2,368
dj_8_MAR	8	50	650	150	1400	2000	9,023
dj_8_FEZ	8	100	150	550	2300	1900	9,023
qaoa_8_SHE	8	1700	3,350	2300	4150	4600	9,023
qaoa_8_KYI	8	2000	3,250	2350	4150	5400	9,023
qft_8_FEZ	8	3300	3,650	3650	3300	4900	9,023
qft_8_TOR	8	3600	3,850	3500	3350	4800	9,023
qnn_8_FEZ	8	2550	3,450	3100	3700	5000	9,023
qnn_8_KYI	8	3150	3,650	3550	3500	4500	9,023
random_8_FEZ	8	2750	3,150	3100	3600	5300	9,023
vqe_8_KYI	8	200	1,050	1100	2700	2700	9,023

yields error distributions that are right-skewed (the agent prefers a few extra shots to a premature, invalid stop), which is the desirable bias in a scientific setting. (iii) *Modularity*: since the policy is learned from the Oracle’s signal rather than hard-coded, swapping the Oracle’s divergence (TVD $\rightarrow$ Hellinger/JS) only requires retraining, not redesign.

Limitations. The agent does not yet outperform the best online heuristic, and its dominant error mode is conservative *undershooting on the hardest, near-budget circuits*, where the dynamic signals are weakest and the correct stop lies far into the budget. Prolonged training induces the opposite pathology – hyper-conservative maximum-budget stops on a few ambiguous easy instances – consistent with overfitting to the asymmetric reward. Both point to the reward shaping and the hard-instance coverage of the training distribution as the primary levers for improvement.

6 Threats to Validity

Construct validity. We score against an a-posteriori Oracle rather than the inaccessible true distribution; this is the same principled reference used by the state of the art [2], and we use the identical δ , budget, and batch size for comparability. *Internal validity.* Quantum sampling is stochastic; we mitigate run-to-run

Table 3. Per-trace shot counts on the *large* circuits ($n \geq 10$): a-posteriori Oracle, the proposed DRL agent (mean over evaluation runs), and the online Inc-* policies / a-priori Weissman bound of Bisicchia et al. [2] at $\delta = 0.10$.

Trace	n	Oracle	DRL (ours)	Inc-TVD	Inc-Hell	Inc-JS	Weissman
dj_10_FEZ	10	100	300	1600	3450	4200	35,600
dj_10_TOR	10	500	1,850	2200	4450	4700	35,600
qaoa_10_SHE	10	6550	4,950	9950	9750	11000	35,600
qaoa_12_MAR	12	7200	5,200	9000	14950	15900	$1.42e+5$
qft_12_TOR	12	15200	7,100	17200	19350	20000	$1.42e+5$
qft_12_FEZ	12	15400	6,450	17150	18150	20000	$1.42e+5$
random_12_FEZ	12	15200	5,800	17100	19250	20000	$1.42e+5$
vqe_12_KYI	12	550	2,800	3400	6150	7500	$1.42e+5$
dj_14_FEZ	14	500	2,200	2450	4850	5200	$5.68e+5$
dj_14_MAR	14	600	2,600	2300	5150	6200	$5.68e+5$
qaoa_14_KYI	14	13350	6,750	13400	20000	20000	$5.68e+5$
qaoa_14_SHE	14	14350	6,550	14600	20000	20000	$5.68e+5$
qft_14_FEZ	14	17750	7,000	17800	20000	20000	$5.68e+5$
qft_14_TOR	14	17750	10,600	17750	20000	20000	$5.68e+5$
qnn_14_FEZ	14	17600	7,550	17700	20000	20000	$5.68e+5$
qnn_14_KYI	14	17700	7,250	17750	20000	20000	$5.68e+5$
random_14_FEZ	14	17700	6,800	17800	20000	20000	$5.68e+5$
vqe_14_KYI	14	2600	4,900	4500	11150	14000	$5.68e+5$

variance through multi-run evaluation and fixed seeds. *External validity.* Experiments use IBM noisy *simulators* from QSimBench rather than live hardware, and target static, non-variational circuits; transfer to real devices and to variational workloads remains future work.

7 Conclusion and Future Work

We presented a Deep Reinforcement Learning approach to adaptive shot allocation that learns, under a strict black-box assumption, when to stop sampling a quantum circuit. Framed as a Markov Decision Process with an eight-feature state and an asymmetric, safety-oriented reward, a Deep Q-Network learns a stopping policy that generalises to unseen circuit families, dramatically outperforms a-priori concentration bounds, and is competitive with purpose-built online heuristics – all without encoding any explicit convergence test. This establishes that black-box shot allocation is learnable.

Future work targets the remaining gap with the best online policy: refining the reward to better handle near-budget hard instances, enriching the state with additional convergence descriptors, exploring distributional and double DQN variants, evaluating mean $\pm$ std over ten runs of the selected agent, and ultimately validating on live quantum hardware and extending the formulation toward variational workloads.

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